Ajay Kumar, Ph.D.

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Professional Summary

Assistant Professor of Physics at Banaras Hindu University with expertise in experimental high-energy physics, specializing in CMS at CERN, Electron-Ion Collider, and heavy-ion collisions. Past research includes Higgs boson searches and silicon sensor development, while current interests focus on advanced detector technologies (GEM, LGAD) and nucleon structure studies, supported by a strong publication record and teaching in particle and computational physics.

Education

- **Ph.D. in Physics (Experimental High Energy Physics)**, University of Delhi, India *10/2010 – 11/2016*
  Thesis: "Search for the SM Higgs Boson in $H \to WW \to l\nu jj$ and Probe of WW Production in Vector Boson Fusion Topology in the CMS Experiment at LHC"
  Supervisor: Prof. Kirti Ranjan
- **M.Sc. in Physics**, University of Delhi, India *07/2008 – 06/2010*
  Percentage: 66.2%, First Division
- **B.Sc. in Physics, Chemistry, Mathematics**, Acharya Narendra Dev College, University of Delhi, India *07/2005 – 06/2008*
  Percentage: 74.44%, First Division; Certificate of Merit for Topping College

Academic Appointments

- **Assistant Professor**, Department of Physics, Institute of Science, Banaras Hindu University, Varanasi, India *07/2020 – Present*
  Teach undergraduate and postgraduate courses in Particle Physics, Mathematical Physics, Computational Physics, Quantum Mechanics, and Electronics. Supervise Nuclear Physics, Optics, and Electronics labs. Serve as academic advisor and contribute to curriculum development.
- **Assistant Professor**, Department of Physics, Sri Aurobindo College, University of Delhi, India *08/2016 – 07/2020*
  Taught courses in Modern Physics, Solid State Physics, Atomic Physics, and Computational Physics. Conducted labs in Thermal Physics, Mechanics, and Electricity.

Research Experience

- **Search for SM Higgs Boson and Vector Boson Studies**, CMS Collaboration, CERN *2011 – 2016*
  Led analysis of Higgs boson decay ($H \to WW \to l\nu jj$), published in *JHEP* (2015). Contributed to WW+WZ cross-section measurements and anomalous gauge coupling studies.
- **CMS Detector Operations and Validation**, CERN *2011 – 2015*
  Served as main contact for alignment parameter error estimation and performed tracker validation, b-tagging scale factor validation, and offline DQM shifts.
- **Radiation-Hard Silicon Sensor Development**, University of Delhi *2016 – 2020*
  Developed data acquisition systems for the Sensor Qualification Center (SQC), certified by CMS in 2019. Contributed to IV/CV measurement software in Python and Lua.

Current Research Interests

- **Simulation and Optimization of GEM Detectors**, Banaras Hindu University *2023 – Present*
Conducted simulation-based research using ANSYS and Garfield++ to optimize electric field configurations and gain behavior, enhancing GEM detector performance.
- **Simulation of Low Gain Avalanche Detectors (LGADs)**, Banaras Hindu University *2023 – Present*
Performed TCAD simulations to optimize LGAD performance for high-radiation environments, enhancing timing resolution for CMS upgrades.
- **Study of Heavy Hadron Production in Au + Au Collisions**, Banaras Hindu University *2024 – Present*
Analyzed heavy hadron production in Au + Au collisions at $\sqrt{s_{NN}} = 200$ GeV to probe Quark-Gluon Plasma properties using PHENIX data.
- **Deeply Virtual Compton Scattering Studies for EIC Physics**, Banaras Hindu University *2022 – Present*
Contributed to simulation studies for the Electron-Ion Collider, focusing on DVCS to probe generalized parton distributions and nucleon structure. Developed analysis frameworks using Python and ROOT to optimize detector requirements for the ATHENA experiment.

Selected Publications

Journal Articles

- J. Adam et al. (2022). ATHENA detector proposal — a totally hermetic electron nucleus apparatus proposed for IP6 at the Electron-Ion Collider. *J. Instrum.*, 17, P10019. DOI:10.1088/1748-0221/17/10/P10019
- CMS Collaboration (2017). Search for anomalous couplings in boosted WW/WZ $\rightarrow l\nu qq$ production in proton-proton collisions at $\sqrt{s} = 8$ TeV. *Phys. Lett. B*, 772, 21–42. DOI:10.1016/j.physletb.2017.06.009
- CMS Collaboration (2015). Search for a Higgs boson in the mass range from 145 to 1000 GeV decaying to a pair of W or Z bosons. *J. High Energy Phys.*, 10, 144. DOI:10.1007/JHEP10(2015)144
- G. Jain, A. Kumar, et al. (2019). Radiation hardness investigation of thin and low resistivity bulk Si detectors. *Nucl. Instrum. Meth. A*, 936, 693–694. DOI:10.1016/j.nima.2018.09.108
- CMS Collaboration (2014). Search for WWγ and WZγ production and constraints on anomalous quartic gauge couplings in pp collisions at $\sqrt{s} = 8$ TeV. *Phys. Rev. D*, 90, 032008. DOI:10.1103/PhysRevD.90.032008
- CMS Collaboration (2014). Alignment of the CMS tracker with LHC and cosmic ray data. *J. Instrum.*, 9, P06009. DOI:10.1088/1748-0221/9/06/P06009
- Full list: INSPIRE-HEP Profile

Conference Papers

- R. Gupta, A. Kumar, et al. (2024). Investigating the effects of acceptor removal mechanism and impact ionization on proton irradiated 300 µm thick LGAD. *Proc. DAE Symp. Nucl. Phys.*, 68.
- G. Jain, A. Kumar, et al. (2019). Development of an automated and programmable characterization system for silicon multi-strip sensors. *Nucl. Instrum. Meth. A*, 936, 663–665. DOI:10.1016/j.nima.2018.10.115
- C. Jain, A. Kumar, et al. (2019). Performance of silicon sensor quality control centre developed at the University of Delhi. *IEEE Proc.*, 978-1-7281-4164-0.

- A. Kumar (2014). Measurement of Electroweak Vector Boson Pair Production in p-p Collision with the CMS Detector at LHC. *Proc. Second Annual LHCP*, CMS-CR-2014/193. arXiv:1409.3414

Teaching Experience

- **Banaras Hindu University** *07/2020 – Present*
  Theory: Particle Physics (PG), Mathematical Physics (PG), Computational Physics (PG), Quantum Mechanics (UG), Electronics (UG).
  Practicals: Nuclear Physics Lab, General and Optics Lab, Electronics Lab.
- **Sri Aurobindo College, University of Delhi** *08/2016 – 07/2020*
  Theory: Elements of Modern Physics, Solid State Physics, Atomic Physics, Computational Physics, Electricity and Magnetism.
  Practicals: Thermal Physics, Wave and Oscillations, Mechanics, Solid State Physics.

Technical Skills

- **Programming and Data Analysis**: Python (NumPy, Pandas, Matplotlib, scikit-learn, Jupyter Notebooks), C++
- **Machine Learning**: Neural networks, boosted decision trees, scikit-learn, TMVA; applied to event classification and detector optimization
- **High-Energy Physics Tools**: ROOT, CMSSW, RooFit, Pythia8, Madgraph, HYDJET++
- **Simulation and Modeling**: ANSYS, Garfield++, Geant4, TCAD (for LGAD simulations)
- **Scientific Computing**: LaTeX, Git

Awards and Honors

- Non-NET Junior Research Fellow, University Grants Commission, India *2010 – 2013*
- Qualified National Eligibility Test (NET), UGC-CSIR, India *2010*
- Certificate of Merit for Topping College, Acharya Narendra Dev College *2009*

Selected Conferences and Workshops

- **Poster Presentation**: Simulation study of GEM detectors, XXVI DAE-BRNS High Energy Physics Symposium, India *2024*
- **Oral Presentation**: TCAD simulation of proton-irradiated LGADs, XXVI DAE-BRNS High Energy Physics Symposium, India *2024*
- **SQC Workshop**, University of Delhi, India *01/2019*
- **AMDFA-2018**, Chandigarh University, India *05/2018*
- Attended multiple HEP workshops and conferences, 2011–2017

Professional Service

- Main Contact Expert, Alignment Parameter Error Estimation, CMS Collaboration *2011 – 2015*
- B-Tagging Scale Factor Validation, CMS Collaboration *2011 – 2015*
- Tracker Validation and DQM Shifts, CMS Collaboration *2012*